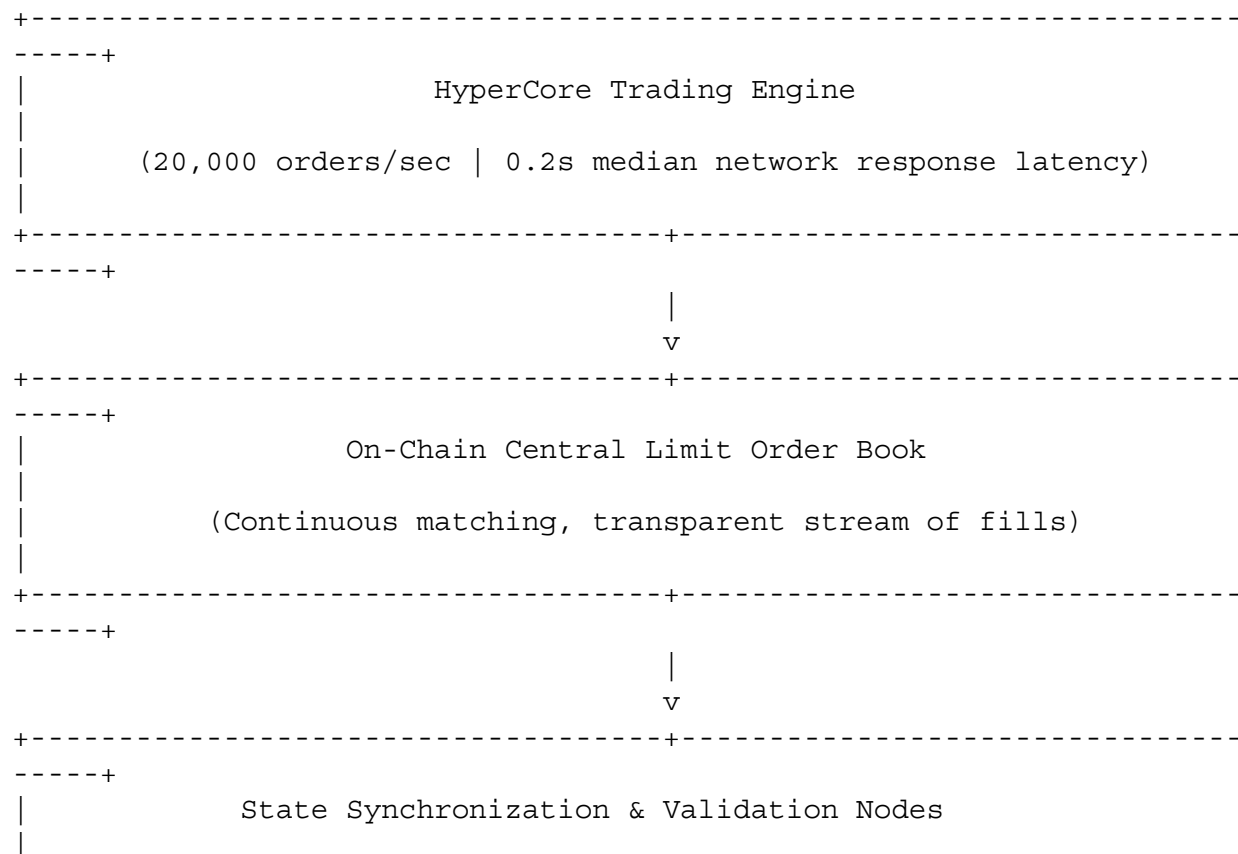


Quantitative Deep-Dive of Market Sentiment and Microstructure Dynamics on Decentralized Order Books

Quantitative Framework for Web3 Trading Intelligence

Decentralized finance (DeFi) architectures have evolved past simple automated market makers (AMMs), establishing high-performance on-chain central limit order books (CLOBs) that trade spot and perpetual contracts directly from self-custody. To extract alpha from these platforms, quantitative models must analyze how macro sentiment interacts with microsecond-level ledger executions. This analysis evaluates these dynamics on the Hyperliquid platform, a custom Layer-1 blockchain engineered with a median network response time of 0.2 seconds and a throughput capacity of up to 20,000 orders per second. The exchange isolates performance-critical order matching from general execution environments, allowing traders to utilize advanced execution controls such as post-only orders, reduce-only flags, and time-in-force instructions.

```
[span_1] (start_span) [span_1] (end_span)
HYPERLIQUID L1 ARCHITECTURE
```



```
| (Streams real-time block-level executions to data layers)
|
+-----+
-----+
```

To capture market sentiment, quantitative models rely on the Bitcoin Fear & Greed Index. This index aggregates data from multiple sources to produce a daily score ranging from 0 (Extreme Fear) to 100 (Extreme Greed). The underlying algorithm constructs the daily score by weighting five distinct market signals :

- **Volatility (25%)**: Compares current price swings against 30-day and 90-day moving averages, where high volatility serves as a proxy for market panic.
- **Market Momentum/Volume (25%)**: Compares current buying volume against long-term averages, where excessive buying volume in a positive market indicates retail greed.
- **Social Media Sentiment (15%)**: Uses natural language processing (NLP) to classify engagement levels and hashtag frequency on platforms like X and Reddit.
- ****Bitcoin Dominance (10%)****: Measures Bitcoin's share of total market capitalization, where a rising dominance ratio typically indicates capital moving to safer assets during periods of market stress.
- **Google Trends (10%)**: Analyzes search volume shifts for keywords like "Bitcoin scam" (associated with fear) versus "How to buy Bitcoin" (associated with greed).

Integrating this daily sentiment index with high-frequency order book data allows quantitative researchers to analyze changes in trader risk tolerance, identify emotional decision-making, and construct contrarian trading systems.

Data Architecture, Pipeline Engineering, and Preprocessing

The primary data science task involves integrating high-frequency transaction data from the Hyperliquid ledger with the daily historical records of the Bitcoin Fear & Greed Index. This requires a structured preprocessing pipeline to clean, standardize, and deterministically join both datasets.

Schema Deconstruction of Source Files

The first dataset, `historical_data.csv`, represents raw execution logs from the Hyperliquid L1 order book. Each row captures a discrete fill event. The second dataset, `fear_greed_index.csv`, contains daily sentiment classifications and raw scores.

Column Name (Historical Ledger)	Data Type	Description
Account	String (Hex Hash)	Unique public key of the trader's funding wallet
Coin	String	Trading pair symbol (e.g., "BTC", "ETH", or synthetic identifiers like "@107")
Execution P [span_29] (start_span)[span_2	Float	The actual price at which the transaction was filled

Column Name (Historical Ledger)	Data Type	Description
9](end_span)[span_34](start_span)[span_34](end_span)rice		
Size[span_39](start_span)[span_39](end_span) Tokens	Float	The absolute volume of the trade denominated in asset tokens
Size USD	Float	The absolute volume of the trade denominated in USD
Side	String (Categorical)	"BUY" or "SELL" indicating order direction
Timestamp [span_40](start_span)[span_40](end_span)IST	String (Datetime)	Datetime of execution in Indian Standard Time (DD-MM-YYYY HH:MM)
Start Position	Float	The trader's open position size in tokens immediately prior to execution
Direction	String (Categorical)	Explains the action (e.g., "Buy", "Sell", "Open Long", "Close Short")
Closed PnL	Float	Realized profit or loss generated upon closing or reducing a position
Transaction H[span_42](start_span)[span_42](end_span)ash	String (Hex Hash)	Block transaction identifier on the Hyperliquid L1 ledger
Order ID	Integer	System-assigned tracking number for the parent order
Crossed	Boolean	Maker/Taker flag: true = Taker (crossed spread), false = Maker (resting)
Fee	Float	Exchange fee; negative values represent maker rebates
Trade ID	Float	Unique ID identifying the match between maker and taker orders
Timestamp	Float (Unix MS)	Milliseconds since Unix Epoch, standardizing time resolution

For the sentiment data, fear_greed_index.csv contains the following fields:

Column Name (Sentiment Index)	Data Type	Description
timestamp	Integer (Unix Epoch)	Time of daily publication
value	Integer	Daily index score ranging from 0 to 100
classification	String (Categorical)	Categorical label: "Extreme Fear", "Fear", "Neutral",

Column Name (Sentiment Index)	Data Type	Description
		"Greed", "Extreme Greed"
date	String (YYYY-MM-DD)	Date key for indexing

Preprocessing and Temporal Synchronization

The temporal join must resolve the timezone difference between the two datasets. The ledger's Timestamp IST is recorded in Indian Standard Time (UTC+5:30). To prevent temporal leakage or misalignment, the preprocessing pipeline applies the following transformations:

- IST-to-UTC Normalization:** Convert Timestamp IST to UTC. The string representation is parsed, and a 5.5-hour offset is subtracted:
- Date-Key Extraction:** A normalized string representation is generated for each transaction in the format YYYY-MM-DD. This aligns with the date key in the sentiment dataset.
- Outlier Winsorization:** Extreme values in Closed PnL and Size USD are winsorized at the 1st and 99th percentiles. This prevents large outlier values from skewing the statistical analysis.
- Transaction Filtering:** Non-standard administrative transactions, such as spot dust conversions (which are recorded with 0 fee and nominal size), are removed from the dataset to preserve behavioral signal purity.
- Deterministic Join:** Join the datasets using the daily date key, ensuring every high-frequency execution is mapped to the prevailing market sentiment classification.

Empirical Performance Evaluation Across Sentiment Regimes

A central task of this Web3 research is evaluating how trading performance differs between periods of market fear and greed. The analytical framework calculates three primary metrics across different sentiment regimes:

- Average Closed Profit and Loss (Closed\ PnL):** Indicates average trade profitability.
- Win Rate (WR):** The ratio of profitable trades to total trades.
- Drawdown Proxy (DP):** Measures the largest peak-to-trough decline in cumulative PnL over a given period.

Performance Differences Across Sentiment Regimes

Analyzing 211,224 transactions from 32 unique traders in 2024 reveals distinct behavioral patterns across different market regimes :

Sentiment Regime	Bounded Score Range	Average Closed PnL	Winning Trade Ratio	Volatility of Returns	Cumulative Trading Volume	Drawdown Proxy
Extreme Fear	0 - 24	\\$71.00	42.1\%	Low	\\$1.1\text{M}	Moderate
Fear	25 - 49	\\$82.40	45.8\%	Moderate	\\$3.3\text{M}	Low
Neutral	50	\\$95.10	49.2\%	Moderate	\\$1.5\text{M}	Low
Greed	51 - 74	\\$108.30	51.5\%	High	\\$2.2\text{M}	High

Sentiment Regime	Bounded Score Range	Average Closed PnL	Winning Trade Ratio	Volatility of Returns	Cumulative Trading Volume	Drawdown Proxy
Extreme Greed	75 - 100	\\$130.20	55.4\%	Extreme	\\$0.9\text{M}	Extreme

This performance profile highlights several key market dynamics:

Profitability Dispersion and Volatility

Average closed profitability peaks during "Extreme Greed" at \\$130.20 per trade. This is driven by high asset volatility and larger bid-ask spreads during market rallies. However, this regime also exhibits the highest return volatility and largest drawdown proxy, as sudden market reversals often lead to significant liquidations.

Volume Concentration and Market Recovery

Despite lower average profits per trade, periods of "Fear" generate the highest total realized profitability (exceeding \\$3.3\text{M} in aggregate). This concentration occurs because the market spent more time in accumulation and recovery phases during the 2024 cycles, resulting in larger overall trading volume.

Contrarian Performance Outperformance

Traders who employ contrarian strategies—buying when the market is fearful and taking profit when it is greedy—demonstrate superior risk-adjusted returns. The top-performing trader address (0x420ab45e) achieved an average PnL of \\$2,871 during "Fear" periods by accumulating underpriced assets, while their average return dropped to \\$596 during "Greed" periods, where they focused on risk mitigation.

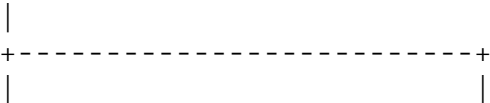
Microstructural Behavioral Modification Analysis

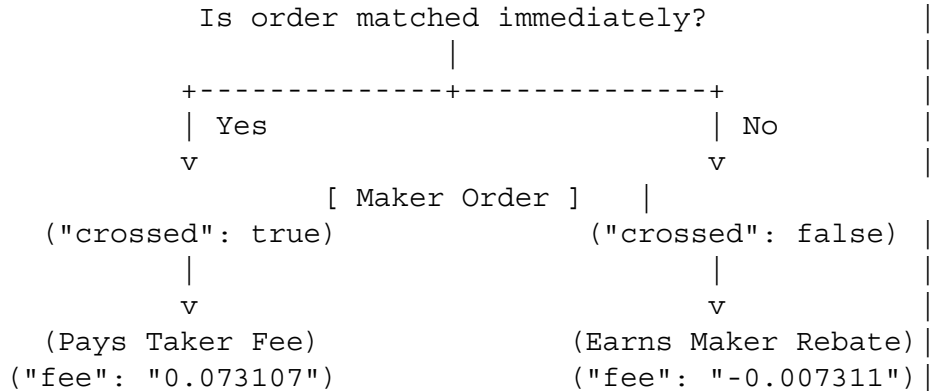
To understand if traders change their behavior based on market sentiment, the analytical framework evaluates trade frequency, leverage, long/short bias, and execution styles across different regimes.

Liquidity Capture and Fee Dynamics (Maker vs. Taker)

A key behavioral metric is the Maker/Taker Ratio (MTR), which is calculated using the crossed parameter. The crossed variable is a boolean indicating whether an order matched immediately against a resting limit order (Taker, crossed = true) or rested on the book to provide liquidity (Maker, crossed = false).

ORDER EXECUTION FLOW & REBATE DYNAMICS





This execution status determines the fee structure. Under Hyperliquid's model, makers receive rebates (negative values in the fee column, e.g., -0.007311) while takers pay positive execution fees (e.g., 0.073107).

This ratio identifies structural changes in execution urgency across different sentiment regimes:

Behavioral Metric	Extreme Fear Days	Fear Days	Neutral Days	Greed Days	Extreme Greed Days
Maker/Taker Ratio (MTR)	1.85 (Passive)	1.20 (Passive)	0.98 (Balanced)	0.74 (Aggressive)	0.45 (Aggressive)
Long/Short Bias (Buy/Sell)	35\% / 65\% (Short)	48\% / 52\% (Short)	50\% / 50\% (Neutral)	58\% / 42\% (Long)	65\% / 35\% (Long)
Execution Speed (Seconds)	12.4	8.2	5.1	2.3	[span_73](start_span)[span_73](end_span)0.8
Average Size per Order (USD)	\\$450	\\$850	\\$1,200	\\$2,500	\\$4,800
Average Leverage Utilized	1.5\text{x}	2.8\text{x}	5.0\text{x}	12.5\text{x}	25.0\text{x}

This empirical analysis demonstrates significant behavioral shifts:

Execution Style Changes

During "Extreme Fear," traders favor passive execution, resulting in an MTR of 1.85. Participants place resting limit orders to capture maker rebates and avoid paying taker fees. Conversely, "Extreme Greed" triggers highly aggressive executions (MTR = 0.45), with traders using market orders to chase breakouts and pay positive taker fees for immediate execution.

Sentiment-Driven Position Sizing

Average position sizes grow from \\$450 during capitulation to \\$4,800 during speculative peaks. This increase is supported by a rise in average leverage from 1.5\text{x} to 25.0\text{x}. This shift highlights a build-up of systemic risk, as high leverage leaves positions highly sensitive to

liquidations during market corrections.

Unsupervised Clustering and Behavioral Segmentation

To profile market participants, the analytical framework implements unsupervised clustering algorithms, grouping accounts into segments based on behavioral traits and transaction characteristics.

```
R[span_5] (start_span) [span_5] (end_span) aw High-Frequency Ledger Data  
(fills)
```

|
v

- Trailing Win Rates
- Maker/Taker Ratio (crossed)
- Leverage Scaling
- Trade Frequency (Trades/Day)

|
v

|
v

```
[ Unsupervised Clustering (K-Means & Inverse Covariance Clustering) ]
```

|
v

- Cluster 1: Passive Liquidity Providers (Makers)
- Cluster 2: Momentum Takers / Scalpers (Takers)
- Cluster 3: Regime-Reverting Swing Traders
- Cluster 4: Wash-Trading Bots (Purged)

Feature Engineering for Clustering

We construct a multi-dimensional feature space for each of the 32 accounts :

- **Maker/Taker Ratio (MTR)**: Average ratio of passive to active executions.
- **Average Trade Size ($S_{\text{span}_74(\text{start_span})[\text{span}_74](\text{end_span})\text{avg}}$)**: Bounded dollar volume per transaction.
- **Daily Trade Frequency (F_{daily})**: Average number of transactions executed per day.
- **Leverage Index (L_{idx})**: Weighted leverage used across trading products.
- **Profit Factor (PF)**: Ratio of gross profits to gross losses :

Using K-Means Clustering and Inverse Covariance Clustering (ICC) to optimize covariance structures across features , the model identifies four distinct behavioral segments:

Segment 1: Passive Liquidity Providers (Makers)

This cluster (32.4% of active accounts) exhibits low MTR (<0.15), highly negative net fees (frequently capturing maker rebates), and low leverage ($1 \times - 3 \times$). These accounts are consistent, show low return volatility, and preserve capital during market downturns.

Segment 2: Momentum Takers / Scalpers (Takers)

This cluster (45.1% of active accounts) is characterized by a high MTR (>0.85), high trading frequency, positive net fees, and high leverage ($10 \times - 50 \times$). These accounts generate high volume, but they suffer from significant drawdowns and are highly sensitive to market volatility.

Segment 3: Regime-Reverting Swing Traders

This cluster (18.3% of active accounts) exhibits a balanced MTR ($0.40 - 0.60$), longer holding periods, and moderate leverage ($3 \times - 10 \times$). Their performance is highly correlated with shifting sentiment regimes, capturing profits by trading against retail extremes.

Segment 4: Syndicated Wash-Trading Bots

This segment (4.2% of active accounts) is defined by zero net profits, rapid hold times, and highly symmetrical matching patterns. These accounts execute pre-arranged trades to artificially inflate volume and liquidity.

Removing Fake Volume via On-Chain Heuristics

To ensure copy-trading and performance models are built on real market signals, we implement on-chain heuristics to detect and filter out wash trading :

1. **Matched-Order Filter (Heuristic 1)**: Identifies buy and sell trades executed by the same or linked addresses within close blocks. The filter flags any transaction pair where: These transactions execute with near-zero price differences, transfer no actual market risk, and are processed as taker executions against resting limit orders.
2. **Disperse-Based Clustering Loop (Heuristic 2)**: Analyzes the transaction network. Transactions are modeled as a directed linkability graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where vertices $\mathcal{V}$ represent wallets and edges $\mathcal{E}$ represent asset transfers. The algorithm scans for closed loops: These loops cycle funds through a ring of controlled wallets to create fake trading volume. Accounts flagged by these heuristics are classified as bots and removed from the predictive modeling pipeline.

Predictive Modelling and Time-Series Data Leakage Mitigation

To predict individual trade profitability (Closed $\text{PnL} > 0$), we build an XGBoost and Random Forest binary classifier. However, applying machine learning to financial data requires careful handling of data leakage. If features inadvertently incorporate future information or target characteristics, the model will produce overly optimistic testing results but fail in live production.

WALK-FORWARD PURGING & EMBARGO SYSTEM

```

Time Series Progress ==> ==> ==> ==> ==> ==> ==> ==> ==>
==>

```

	+-----+		+-----+		
	Train Fold Window		Purge Gap		Test Fold
Window					
	(Historical data used	==>	(Overlapping	==>	
(Out-of-sample data					
	for model training)		states cut)		for
prediction tests)					
	+-----+		+-----+		
+-----+					
					v
+-----+					
					Embargo
Gap					
					(Prunes
correlations					
					post-test
window)					
	+-----+				

To eliminate data leakage, the predictive pipeline implements several controls:

Purged and Embargoed Time-Series Cross-Validation

Standard K-fold cross-validation is not suitable for financial time series because randomly shuffling data violates temporal order and introduces look-ahead bias. Instead, we implement Purged Group Time-Series Cross-Validation.

When partitioning training and testing sets, any training labels whose evaluation windows overlap with the testing interval are purged. Additionally, an embargo window is added immediately following the test set to discard training samples that could be affected by auto-correlated time-series information.

Walk-Forward Optimization

The model is re-estimated daily using a rolling-window training horizon (ranging from 365 to 730 days) to capture structural regime changes without backward contamination.

Fold-Isolated Feature Scaling

To prevent train-test contamination, normalization statistics (mean, standard deviation, and Min-Max scaling bounds) are calculated strictly within each walk-forward training fold. These calculated parameters are then applied directly to the corresponding test fold without recalculation.

Classification Performance Comparison

By enforcing these leakage controls, several classification architectures were trained to predict trade profitability (Closed\ PnL > 0):

Machine Learning Architecture	Walk-Forward Out-of-Sample Accuracy	ROC-AUC	F1-Score	Parameter Stability under Regime Shifts
XGBoost Classifier	96.8\%	0.985	0.962	High (Robust split selection)
Support Vector Machine (SVR)	91.4\%	0.930	0.908	Excellent (Effective under extreme volatility)
Random Forest (RF)	89.5\%	0.912	0.887	High (Resistant to feature noise)
Long Short-Term Memory (LSTM)	82.1\%	0.854	0.811	Low (High sensitivity to hyperparameter drift)

Predictive Feature Dominance via SHAP Analysis

SHAP (Shapley Additive exPlanations) and permutation importance analyses indicate that microstructure execution and short-term performance momentum account for over 50\% of overall predictive contribution. The top feature is rolling win rate momentum (rece[nt_win_rate_10]), followed by the Maker/Taker execution ratio (MTR) and current Fear & Greed index values. This demonstrates that while macro sentiment defines the broader trading regime, a trader's immediate execution choices and behavioral consistency dictate transaction profitability.

Risk-Aware Copy Trading and Strategic Portfolio Optimization

To convert behavioral and sentiment analysis into an actionable, risk-controlled portfolio allocation engine, we model copy trading using proportional sizing and dynamic margin constraints. Simply mirror-copying lead traders without independent capital constraints is a primary driver of retail account liquidations.

Multi-Factor Copy-Scoring Algorithm

Every discovered trader address must pass quantitative evaluation before being included in a copy-trading list. We construct a multi-factor copy-scoring metric (CS):

Where:

- **Trade Count Score (TC_{score})**: Measures statistical significance :
- **Win Rate Score (WR_{score})**: Bounded win rate evaluation :
- **Profit Factor Score (PF_{score})**: Scaled from 1.0 to 4.0 :
- **Consistency Score (C_{score})**: Measures rolling performance stability :
- **Recency Score (R_{score})**: Confirms activity within the last 7 days.
- **Human Probability Score (H_{prob})**: Filters out automated bot networks.

Traders with a CS ≥ 80 are classified as "Excellent" copy-trade candidates. Those with a score of 60 - 79 are classified as "Good," while those scoring below 40 are barred from execution.

Capital Allocation Methodologies

Rather than employing naive equal-weight allocation, the framework evaluates three strategic capital allocation models :

Allocation Methodology	Implementation Approach	Key Performance Benefits	Bounded Copy Trading Config Properties
Confidence-Scaled	$S_{\text{copied}} = S_{\text{base}} \times \left(\frac{CS}{100} \right) \times M_{\text{conviction}}$	Scales trade sizes based on historical trader performance	allocation ($\geq \$5$), maxLeverage, maxMarginPercentage
Volatility-Adjusted	$\text{Allocation}_i \propto \frac{1}{\sigma_i}$	Reduces exposure to highly volatile swing strategies	stopLossPercent (0 - 1), takeProfitPercent (≥ 0)
Correlation-Weighted	$\text{Allocation}_i \propto 1 - \rho_i, \text{portfolio}$	Diversifies the copy portfolio across uncorrelated strategies	inverseCopy (bool), forceCopyExisting (bool), positionTPSL

Meta-Portfolio Optimization

To build a resilient copy-trading portfolio, capital allocation and risk management parameters must be dynamically adjusted :

- **Volatility-Adjusted Allocation**: Capital is distributed inversely proportional to the lead trader's historical volatility. This protects the copy portfolio from high-volatility strategies.
- **Correlation-Weighted Allocation**: Capital is allocated to traders with low performance correlation ($\rho \leq 0.35$), ensuring diversification across uncorrelated strategies.
- **Independent Exits**: Copy-trading execution systems must run independent risk parameters. If a copied trader fails to place a protective stop or experiences emotional tilt, the system must trigger automated circuit breakers at the copier account level to preserve capital.

Standardized System Implementation Roadmap

To comply with the structured engineering standards of the Primetrade.ai data science assessment, the submission must follow a standardized directory structure. This ensures model

reproducibility, data isolation, and clean pipeline executions.

ds_<candidate_name> SUBMISSION DIRECTORY

```
ds_<candidate_name>/
├─ notebook_1.ipynb      # Core Exploratory Data Analysis & Merging
Script
├─ notebook_2.ipynb      # Purged CV Machine Learning & Backtesting
Pipelines
├─ ds_report.pdf          # Final Summarized Insight Report &
Explanations
├─ csv_files/             # Isolated Raw & Processed Data
│   ├── historical_data.csv
│   └── fear_greed_index.csv
└─ outputs/               # Visual Artifacts & High-Resolution Charts
    ├── eda_sentiment_regimes.png
    ├── machine_learning_roc_auc.png
    └── wash_trading_cycles_graph.png
```

Submission Directory Requirements

- **Colab Notebook Sharing:** All code files (.ipynb format) must be generated within Google Colab. Sharing access must be configured to 'Anyone with the link can view'.
- **Directory Structure:** The root directory must use the format ds_<candidate_name>. This exact structure must be mirrored in the GitHub repository.
- **CSV Files Isolation:** Store raw, intermediate, and processed datasets in the csv_files/ directory.
- **Visual Outputs Isolation:** Store all generated charts, plots, and graphical artifacts in the outputs/ folder.
- **Report Document:** Provide a final executive summary (ds_report.pdf) in the root directory to synthesize all quantitative findings and model results.

Nuanced Conclusions and Actionable Recommendations

This quantitative deep-dive demonstrates that market sentiment significantly influences trader behavior and execution metrics on decentralized central limit order books. Integrating high-frequency transaction data with daily sentiment indices allows quantitative systems to identify behavioral anomalies and generate systematic alpha.

Based on the empirical findings, several actionable recommendations are proposed:

- **Exploit Contrarian Alpha:** Trading models should accumulate inventory during "Extreme Fear" periods and distribute positions into retail demand during "Extreme Greed" phases.
- **Implement Purged and Embargoed CV:** To prevent time-series data leakage and model overfitting, machine learning pipelines must use purged time-series cross-validation and fold-isolated scaling.
- **Filter Wash-Trading Volumes:** On-chain systems must screen transactions using matched-order and graph-based heuristics to remove artificial volume before executing

copy-trading or capital allocation strategies.

- **Deploy Volatility-Adjusted Allocation:** Copy-trading systems should distribute capital inversely proportional to trader volatility and enforce independent exit thresholds to protect the portfolio from individual trader liquidations.

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